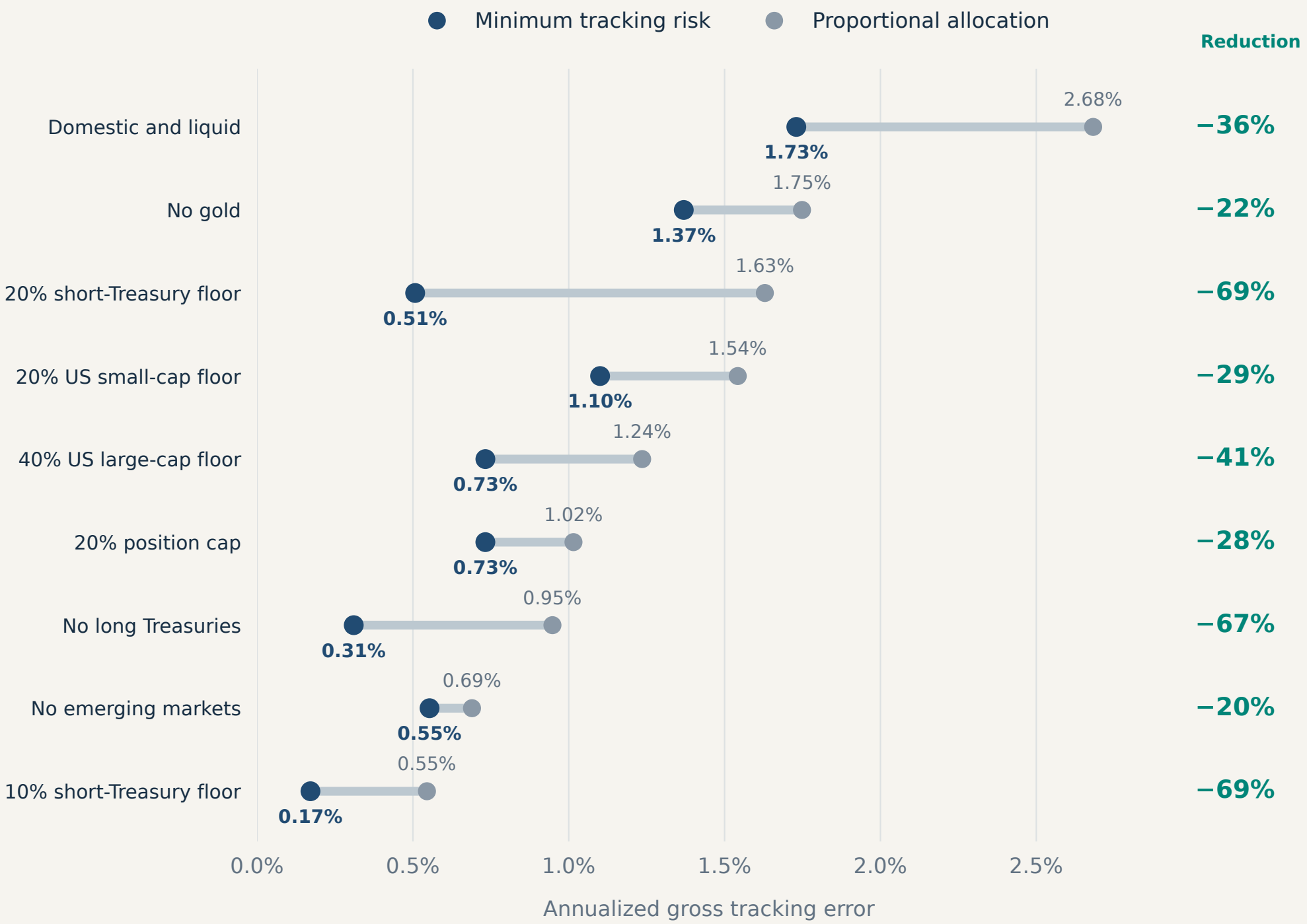
Closer to the model in all 9 constrained cases

Risk-aware allocation reduced realized tracking error by 20–69% versus proportional allocation.
Tracking error measures variability of returns relative to the policy model; lower means closer tracking.

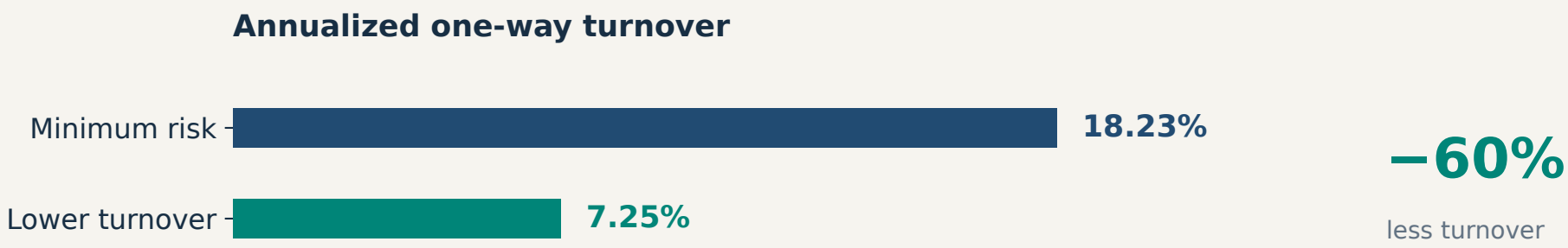


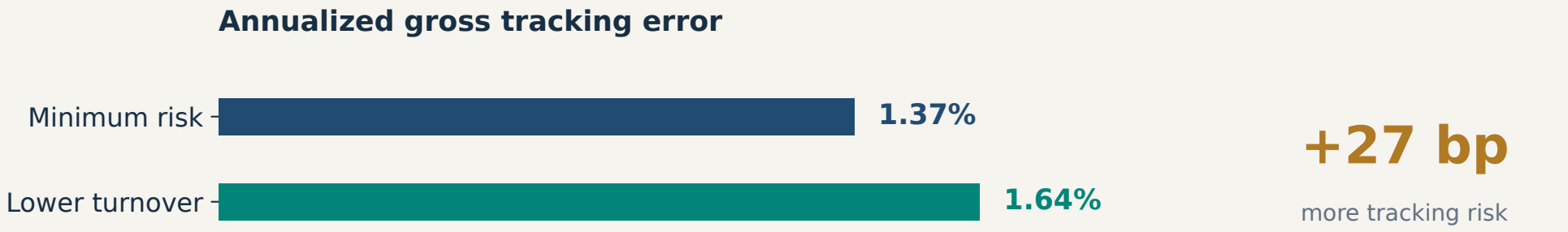
Same restrictions and monthly execution timing for both methods. Full constrained-scenario set shown.
“Domestic and liquid” excludes EFA/EEM and sets a 15% SHY floor. Tracking risk is not expected return.

SOURCE WRDS / CRSP CIZ | 2014–2025 evaluation | Hypothetical ETF accounts

60% less turnover. About $110/year saved.

No-gold account, per $1 million: lower-turnover allocation versus minimum tracking risk.
The cost benefit is small at the assumed 5 bp per traded dollar; tracking risk rises by 27 bp.







Added TE: 27.0 bp; paired 95% block-bootstrap interval 20.0–35.6 bp (1,000 resamples).
$110/year is the annualized cost-rate difference scaled to a constant $1m NAV, not a return forecast.

SOURCE WRDS / CRSP CIZ | 2014–2025 evaluation | Hypothetical ETF accounts

Closest tracking did not mean highest return

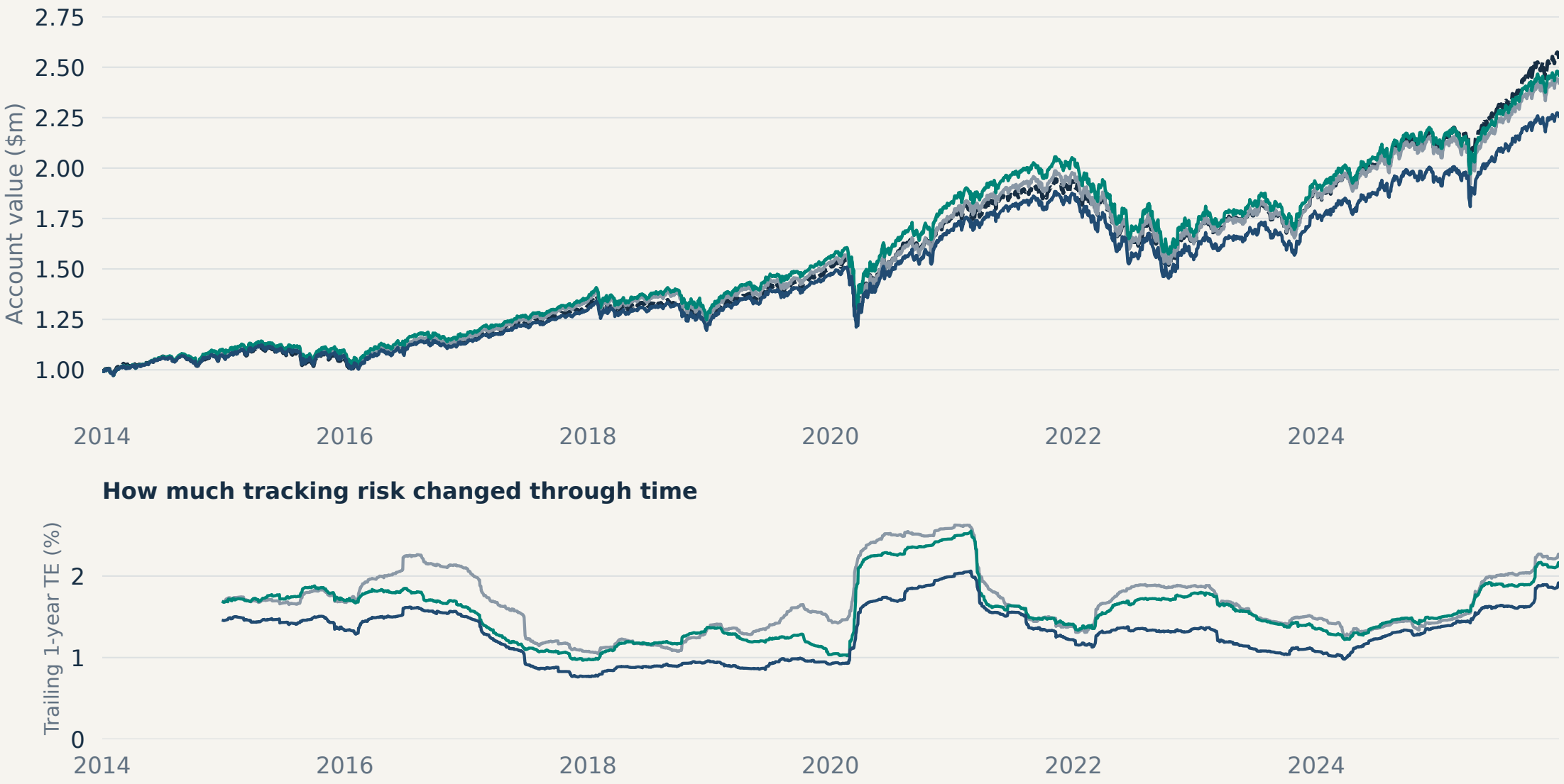
No-gold accounts, starting at $1 million. The policy benchmark includes a 10% gold allocation.
Differences reflect exposure choices—not evidence of alpha or a forecast of future performance.

Policy benchmark
$2.54m
8.10% annualized return

Lower turnover
$2.46m
7.81% annualized return

Proportional
$2.42m
7.66% annualized return

Minimum risk
$2.26m
7.03% annualized return

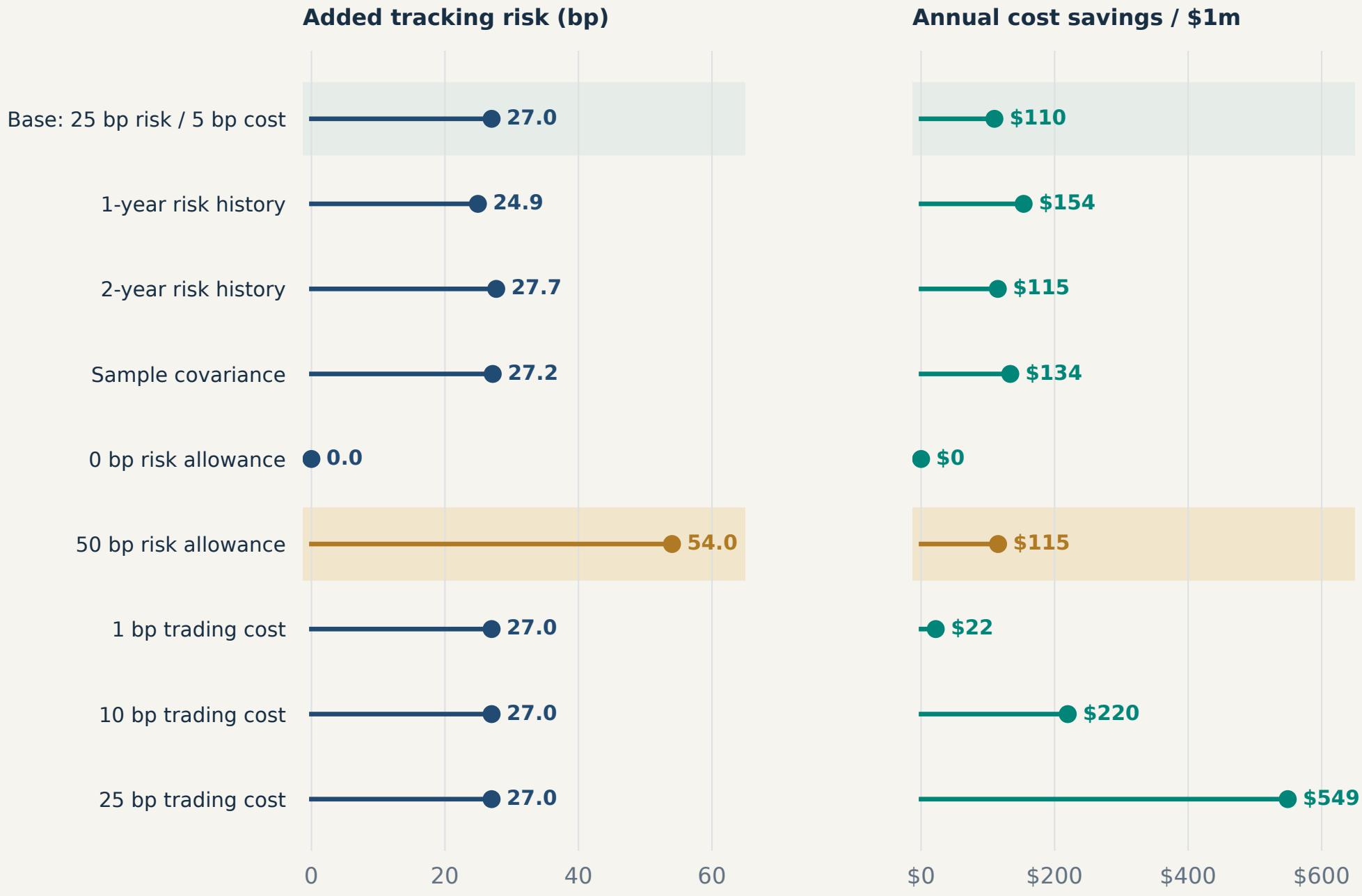


Portfolio wealth and returns are net of modeled trading costs; the policy benchmark is frictionless.
Trailing TE uses 252 daily observations, before trading costs. Initial positions transition at the first close.

SOURCE WRDS / CRSP CIZ | 2014–2025 evaluation | Hypothetical ETF accounts

Doubling the risk budget barely saves more

No-gold account: base specification plus all eight sensitivity runs. Each row is a full simulation.
A 50 bp allowance doubles added realized tracking risk, but saves only ~$5/year more per $1m.



Base: 756-day Ledoit–Wolf risk estimate; estimated TE allowance of 25 bp; 5 bp per traded dollar.
Rows change one assumption. Savings scale annual cost-rate differences to $1m, not actual fixed dollar bills.

SOURCE WRDS / CRSP CIZ | 2014–2025 evaluation | Hypothetical ETF accounts